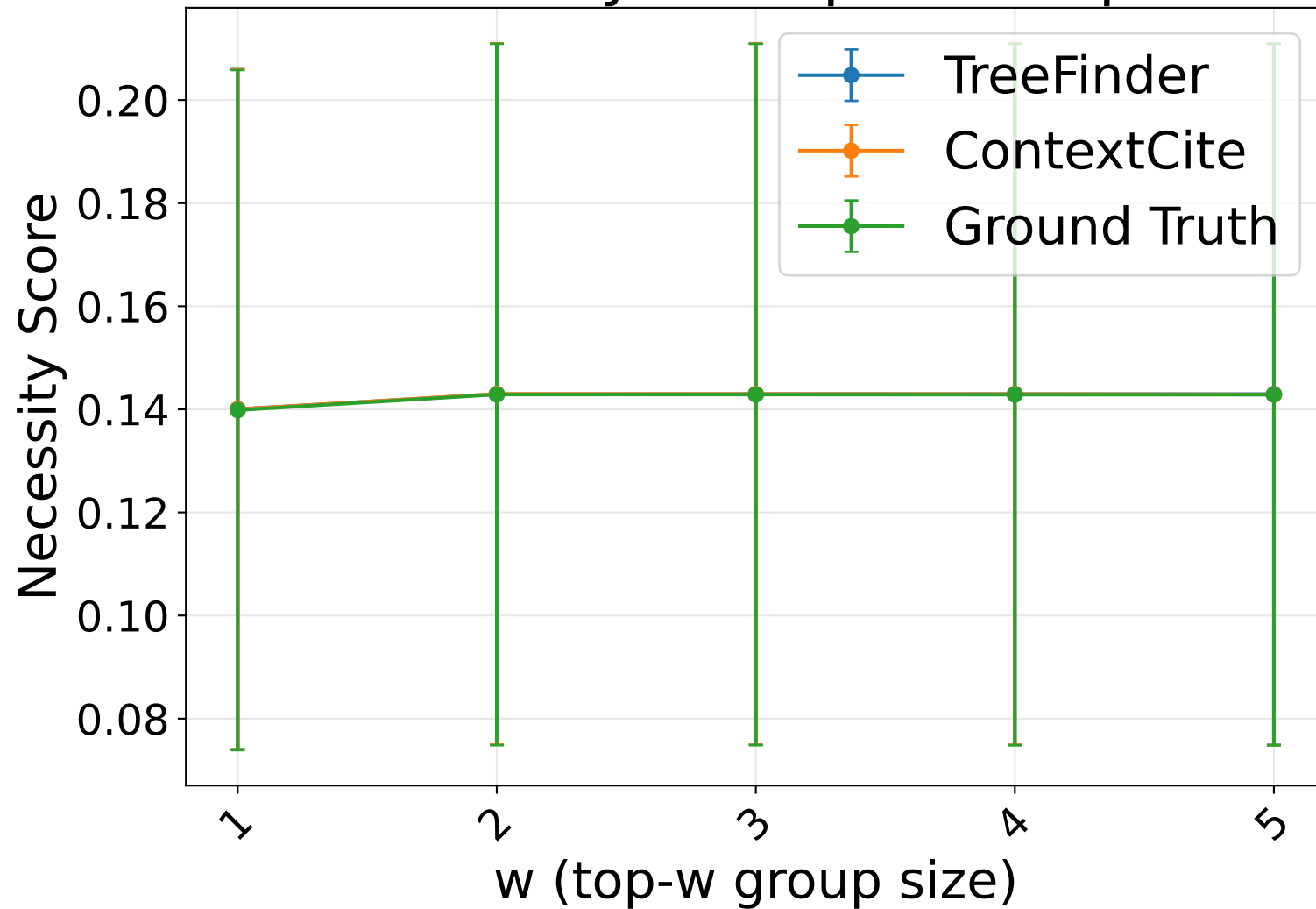


Necessity for Top-w Groups



Sufficiency for Top-w Groups

